

**SUBJECT OUTLINE****Subject code and name**

BDA601 Big Data and Analytics

**SECTION 1 – GENERAL INFORMATION****1.1 Administrative details**

| Associated award/s                                                                                             | Duration                                             | Level |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------|
| Master of Software Engineering (Cloud Computing)<br>Master of Software Engineering (Cloud Computing, Advanced) | 1 Trimester<br>(12 weeks, or intensive over 6 weeks) | 600   |

**1.2 Core or elective subject**

- ☐ core subject (within associated awards)  
☐ elective subject (within associated award/s, and in other approved awards)  
☒ other (please specify below):

Specialisation

**1.3 Subject weighting**

| Subject credit points | Total course credit points                                       |
|-----------------------|------------------------------------------------------------------|
| 10                    | Master: 120 credit points<br>Master, Advanced: 160 credit points |

**1.4 Student workload**

| Duration         | Facilitated study h/p/w | Personal study h/p/w | Total study h/p/w |
|------------------|-------------------------|----------------------|-------------------|
| 12-week duration | 3                       | 7                    | 10                |
| 6-week duration  | 6                       | 14                   | 20                |

**1.5 Delivery mode**

- ☒ Face-to-face  
☒ Online  
☐ Intensive (e.g. residential, or summer school, weekend workshop - details below)

- ☐ Work-integrated learning activity
- ☒ Hybrid
- ☒ Full-time
- ☒ Part-time
- ☐ Other (please specify)

#### 1.6 Pre-requisites and co-requisites

- ☒ Yes ☐ No

If YES, provide details of the prerequisite or co-requisite requirements below.

CCF501 Cloud Computing Fundamentals (pre-requisite)

#### 1.7 Specialist facilities and/or equipment requirements

- ☒ Yes ☐ No

If YES, provide details of specialist facilities and/or equipment below.

Computer/laptop and relevant specialist software.

## SECTION 2 – ACADEMIC DETAILS

### 2.1 Subject Descriptor

This subject teaches students techniques and best practices in collecting, storing, extracting, cleaning, manipulating, visualising and analysing , useful information from large, structured, semi-structured, or unstructured data sets, which are critical for organisational decisions. Students will also learn applications of big data analytics across industry sectors such as segmentation and prediction, recommender systems and targeted marketing, sentiment analysis, operational analytics, and Big Data for social good.

### 2.2 Subject Learning Outcomes

| Subject Learning Outcomes                                                                                 | Mapping to Course Learning Outcomes                                 |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| a) Explain and evaluate the V's of Big Data (Volume, Velocity, Variety, Veracity, Valence, and Value).    | MSE (CC): CLO 7, CLO 8<br>MSE (CC, Adv): CLO 7, CLO 8               |
| b) Identify best practices in data collection and storage, including data security and privacy principles | MSE (CC): CLO 1, CLO 3, CLO 7<br>MSE (CC, Adv): CLO 1, CLO 3, CLO 7 |
| c) Synthesis data science principles and apply to the cleaning, manipulation and visualisation of data    | MSE (CC): CLO1, CLO3, CLO 7<br>MSE (CC, Adv): CLO1, CLO3, CLO 7     |
| d) Design analytical models based on given problems.                                                      | MSE (CC): CLO3, CLO 7<br>MSE (CC, Adv): CLO 3, CLO 7                |
| e) Effectively report and communicate findings to an appropriate audience.                                | MSE (CC): CLO2, CLO4, CLO5<br>MSE (CC, Adv): CLO2, CLO4             |

\* The full list of Course Learning Outcomes and their descriptors is provided in Section 2.6 below.

### 2.3 Assessment Tasks

#### 12-week duration

| No. | Type and Description                                                   | Assessment due       | Weighting | Subject Learning Outcome(s) assessed |
|-----|------------------------------------------------------------------------|----------------------|-----------|--------------------------------------|
| 1.  | <b>Title:</b> Design Data Pipeline<br><b>Delivery type:</b> Individual | Module 4<br>(Week 4) | 30%       | a), b),<br>e)                        |

|    |                                                                                                                                                                                  |                                      |     |            |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----|------------|
|    | <b>Outcome:</b> Report<br><b>Volume of assessment:</b> 1500 words +/- 10% with supporting images                                                                                 |                                      |     |            |
| 2. | <b>Title:</b> Visualisation and Model development<br><b>Delivery type:</b> Individual<br><b>Outcome:</b> Source code + Report<br><b>Volume of Assessment:</b> 1000 words +/- 10% | Module 8<br>(Week 8)                 | 30% | c), d), e) |
| 3. | <b>Title:</b> Model Evaluation<br><b>Delivery type:</b> Individual<br><b>Outcome:</b> Source code & Presentation<br><b>Volume of Assessment:</b> 7-10 mins                       | Wednesday,<br>Module 12<br>(Week 12) | 40% | c), d), e) |

**6-week duration**

| No. | Type and Description                                                                                                                                                                                    | Assessment due                      | Weighting | Subject Learning Outcome(s) assessed |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------|--------------------------------------|
| 1.  | <b>Title:</b> Design Data Pipeline<br><b>Delivery type:</b> Individual<br><b>Outcome:</b> Report Mode:<br><b>Volume of Assessment:</b> 1500 words +/- 10% with supporting images                        | Module 4<br>(Week 2)                | 30%       | a), b), e)                           |
| 2.  | <b>Title:</b> Visualisation and Model Development<br><b>Delivery type:</b> Individual<br><b>Outcome:</b> Source code + report<br><b>Volume of Assessment:</b> Source Code and Report 1000 words +/- 10% | Module 8<br>(Week 4)                | 30%       | c), d), e)                           |
| 3.  | <b>Title:</b> Model Evaluation<br><b>Delivery type:</b> Individual<br><b>Outcome:</b> Source code & Presentation<br><b>Volume of Assessment:</b> 7 – 10 minutes                                         | Wednesday,<br>Module 12<br>(Week 6) | 40%       | c), d), e)                           |

## 2.4 Prescribed texts

These are core texts that are fundamental to meeting subject outcomes. Students may choose to purchase a copy. A limited number of these texts will be available for loan from the campus library.

There are no prescribed texts for this subject.

## 2.5 Suggested readings and resources

These are supplementary readings and resources that support subject knowledge. These may be available from the campus library. Refer to *BDA601 Big Data and Analytics* suggested readings and resources for the list of readings and resources.

**Policies and Procedures, and related application forms, are available via the Torrens University Australia website.**

## 2.6 Course Learning Outcomes

| Course Learning Outcome Code | Course Title: Master of Software Engineering (Cloud Computing)<br>Course Learning Outcome Descriptor                                                                                                  |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLO 1                        | Analyse and integrate cognitive and technical skills from the software engineering body of knowledge to tackle complex commercial software engineering problems with innovation and ethical practice. |
| CLO 2                        | Critically analyse problem sets to synthesise acute software engineering solutions and communicate recommendations, conclusions and professional decisions to both expert and novice audiences.       |
| CLO 3                        | Master and synthesise cognitive, technical and creative skills from the field of cloud computing necessary to effectively & efficiently solve or investigate problem sets.                            |
| CLO 4                        | Demonstrate sound judgement, non-partisanship and accountability as part of leadership role in a team to achieve effective planning, management, execution and communication.                         |
| CLO 5                        | Communicate technical solutions, recommendations and workarounds efficiently to a variety of audiences, including diverse multi-cultural groups.                                                      |
| CLO 6                        | Synthesise and abstract Cloud Computing concepts, devops basics and industry best practices to solve varied commercial problems.                                                                      |
| CLO 7                        | Possess a comprehensive and in-depth knowledge of Cloud Computing and tools that includes understanding of recent developments.                                                                       |
| CLO 8                        | Research Cloud Computing challenges including diversity and ethical practice to generate appropriate strategic solutions.                                                                             |

| Course Learning Outcome Code | Course Title: Master of Software Engineering (Cloud Computing, Advanced)<br>Course Learning Outcome Descriptor                                                                                        |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLO 1                        | Analyse and integrate cognitive and technical skills from the software engineering body of knowledge to tackle complex commercial software engineering problems with innovation and ethical practice. |
| CLO 2                        | Critically analyse problem sets to synthesise acute software engineering solutions and communicate recommendations, conclusions and professional decisions to both expert and novice audiences.       |
| CLO 3                        | Master and synthesise cognitive, technical and creative skills from the field of cloud computing necessary to effectively & efficiently solve or investigate problem sets.                            |
| CLO 4                        | Demonstrate sound judgement, non-partisanship and accountability as part of leadership role in a team to achieve effective planning, management, execution and communication.                         |
| CLO 5                        | Communicate technically to present coherent and sustained arguments and to disseminate research that makes a contribution to knowledge.                                                               |
| CLO 6                        | Synthesise and abstract Cloud Computing concepts, devops basics and industry best practices to solve varied commercial problems.                                                                      |
| CLO 7                        | Possess a comprehensive and in-depth knowledge of Cloud Computing and tools that includes understanding of recent developments.                                                                       |
| CLO 8                        | Research Cloud Computing challenges including diversity and ethical practice to generate appropriate strategic solutions.                                                                             |